

# ACME Salary Management Platform (MVP)

Author: Biswajit Sen

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## 1. Background & Problem Statement

ACME manages compensation data for approximately 10,000 employees distributed across multiple countries. Currently, HR teams maintain salary information using fragmented, version-uncontrolled spreadsheets. As the organization grows, this setup introduces compounding operating risks:

- **Data Fragmentation:** Salary records are scattered across multiple systems, emails, and local drives, making it difficult to maintain a definitive "source of truth."
- **Zero Auditability:** Historical compensation changes are completely untraceable, significantly heightening compliance, legal, and operational risks.
- **Reporting Bottlenecks:** Answering basic corporate or leadership requests (such as department averages or regional payroll spend) requires hours of manual cross-referencing and calculation.

## 2. High-Level Strategic Goals

The core objective is to deliver a secure, centralized, web-based platform that entirely supersedes spreadsheet workflows. The product must cleanly manage historical records over time, provide instantaneous organizational analytics, and effortlessly scale to handle over 10,000 baseline records without performance degradation.

## 3. Target User & Success Criteria

- **Primary Persona:** HR Manager (responsible for processing record updates, running performance reviews, and compiling regular reports for leadership).
- **Success Metrics:**
  - Locate or filter any individual employee file within **less than 2 seconds**.
  - Process compensation revisions or historical legacy migrations with **zero data corruption** or destructive overrides.
  - Generate instant high-level analytical answers natively without reliance on external tools.

## 4. MVP Core Scope & Features

### 4.1. Employee Directory & Profile

Provides a consolidated corporate layout equipped with a paginated data grid. Features include global server-side search (by Name or Employee ID) and multi-parameter filtering (by Country, Department, and Job Title). To safely scale up to 10,000 rows, the API strictly enforces pagination limits (e.g., 50 per page) coupled with list virtualization on the client interface to protect browser memory performance.

## 4.2. Compensation Profile Page

When an HR Manager clicks an employee row in the directory, the system routes to a dedicated **Compensation Profile View**. This layout is partitioned into three key modules:

- **Employee Information:** Basic demographical and corporate summary cards (ID, Name, Department, Job Title, Country).
- **Current Compensation:** High-visibility metric highlights showing current Base Salary, active Currency tier, and Last Updated parameters.
- **Salary Timeline:** A clean chronological grid charting the historical progression sequence, tracing metadata such as Previous Salary → New Salary, Change Reason, Operator Identity, and corresponding effective markers.

## 4.3. Compensation Management (Append-Only Audit Trail)

Tracks structural compensation fields: Employee ID, Base Salary, Currency, Effective Date, Reason, Changed By, and Notes. The platform strictly forbids destructive updates to existing rows to maintain historical alignment. Every configuration shift writes a fresh event instance to the database, capturing explicit operational contexts:

- **Enforced Reason Enums:** Annual Increment, Promotion, Market Adjustment, Correction, New Hire.
- **Operational Tracking:** Identifies the exact actor (e.g., HR Manager) and appends contextual field notes for downstream auditing procedures.

### The Absolute Currency Rule

To avoid introducing complex live conversion third-party integrations in the MVP, **all dashboard metrics and salary visualizations are isolated and aggregated strictly by individual Currency/Country code**. Blending disparate currencies together is blocked at the interface level to ensure data integrity.

## 4.4. Automated Analytics Dashboard

Renders real-time operational visual blocks including: Total Headcount, Total Payroll Cost (grouped purely by currency), Average & Median Salaries (grouped by department/currency), and a Top-10 Corporate Earners listing to instantly call out edge outliers.

## 4.5. Fail-Safe Spreadsheet Import (.xlsx)

Enables swift legacy data onboarding by processing standard Excel schemas using the unique Employee ID as an idempotent upsert key. To guarantee absolute database integrity, the platform mandates an **"All-or-Nothing" transactional dry-run gate** mapping to explicit validation constraints:

- **Validation Pipeline Constraints:** The parser strictly checks and rejects files displaying: Duplicate Employee IDs within the sheets, Invalid Currency Codes (non-ISO standard), Negative Base Salaries, Missing Employee IDs, or Empty Effective Dates.
- **The Stoplight Preview Grid:** If 100% of rows are valid, the platform displays an "All Clear" summary and enables a "Confirm Import" execution button. If any violations occur, the active database import is fully locked out, forcing an analytical error view pinning rows to their exact cell failures.

## 4.6. Intent-Driven AI Insights

Empowers HR managers to ask standard computational natural language questions (e.g., *"What is the average salary in Engineering?"*) and transforms them into quick data outputs. To eliminate risk, the system utilizes an internal safety pipeline rather than allowing unstructured SQL generation:

*Natural Language Query → Intent Detection Parsing Layer → Whitelisted Analytics Service Execution → Read-Only Local DB*

Input strings map cleanly to safe pre-written parameterized analytical routines (e.g., Average Department Pay or Top Earners extraction), rendering the mechanism completely immune to AI-driven SQL Injection attacks.

## 5. Out of Scope (Strict Exclusions)

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The following categories are purposefully excluded from the MVP baseline to safeguard the launch timeline:

- **Direct Payroll Operations:** Printing payslips, banking integrations, automated transfers, or tax tranche withholdings.
- **Core HRIS Features:** Timesheets, vacation management, recruitment pipelines, and self-service portals.
- **Granular RBAC:** Multi-stage approval pathways or conditional editing fields (a single administrative HR Manager role is assumed).

## 6. Technical & Non-Functional Requirements

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- **Database Layer:** SQLite database utilizing an explicit relational 2-table model (employees and compensation\_history).
- **Performance Budget:** Maximum page load budget of 2.0 seconds for the analytical summary matrix and queries when loaded with a 10,000-row dataset.
- **Test Infrastructure:** Isolation between backend endpoints and front-end rendering. Core unit tests must cover legacy import parsers and append-only immutability rules.

## 7. Core MVP Database Schema Architecture

| Table Name           | Column Name    | Data Type | Validation / Structural Constraints                                                   |
|----------------------|----------------|-----------|---------------------------------------------------------------------------------------|
| employees            | id             | VARCHAR   | PRIMARY KEY (Unique Employee Identifier)                                              |
|                      | full_name      | VARCHAR   | NOT NULL                                                                              |
|                      | department     | VARCHAR   | INDEXED (For speedy filtering)                                                        |
|                      | job_title      | VARCHAR   | NOT NULL                                                                              |
|                      | country        | VARCHAR   | INDEXED (For speedy filtering)                                                        |
| compensation_history | id             | INTEGER   | PRIMARY KEY AUTOINCREMENT                                                             |
|                      | employee_id    | VARCHAR   | FOREIGN KEY → employees.id                                                            |
|                      | base_salary    | REAL      | NOT NULL (Must evaluate > 0)                                                          |
|                      | currency       | VARCHAR   | NOT NULL (3-Letter Standard ISO Code)                                                 |
|                      | effective_date | DATE      | NOT NULL                                                                              |
|                      | reason         | VARCHAR   | NOT NULL (Enum: Annual Increment, Promotion, Market Adjustment, Correction, New Hire) |
|                      | changed_by     | VARCHAR   | NOT NULL (Identifies administrative user processing change)                           |
|                      | notes          | TEXT      | NULLABLE (Optional text for extra operational context)                                |
|                      | created_at     | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP                                                             |